

JEE MAIN REVISION MASTER: COMMUNICATION SYSTEMS OUT OF SYLLABUS

Subject: Physics | Target Scope: JEE Main Previous Years | Empirical Base: 98 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Communication Systems - Core Application	98 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Communication Systems - Core Application (Very High Yield — 98 PYQs)

Core Principle: Key problem pattern in Communication Systems.

■ ■ **Caution / JEE Pitfall:** Watch for algebraic and sign errors.

■ **Speed Trick:** Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "The height of transmitting antenna is 180m and the height of the receiving antenna is 245m . The maximum distance between them for satis..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (MCQ): "In satellite communication, the uplink frequency band used is :..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (MCQ): "The power radiated from a linear antenna of length l is proportional to (Given, λ = Wavelength of wave):..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 15th April Morning Shift (Q#1)

The height of transmitting antenna is 180m and the height of the receiving antenna is 245m . The maximum distance between them for satisfactory communication in line of sight will be : (given $R=6400\text{km}$)

- (A) 104km
- (B) 56km
- (C) 48km
- (D) 96km

Correct Answer: None

Solution Analysis: $d_{\text{(max)}} = \sqrt{2R h_{\text{(t)}}} + \sqrt{2R h_{\text{(r)}}}$ where $d_{\text{(max)}}$ is the maximum distance between the antennas, R is the radius of the earth, $h_{\text{(t)}}$ is the height of the transmitting antenna above the earth's surface, and $h_{\text{(r)}}$ is the height of the receiving antenna above the earth's surface. Substituting the given values, we get: $d_{\text{(max)}} = \sqrt{2 \times 6400\text{km} \times 180\text{m}} + \sqrt{2 \times 6400\text{km} \times 245\text{m}} = 104\text{km}$ Therefore, the maximum distance between the transmitting and receiving antennas for line of sight communication is 104km .